

Principal Component Analysis

Modelització Estadística

Grau en Intel·ligència Artificial

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What is Principal Component Analysis?

Principal Component Analysis (PCA)

PCA is a technique for:

- **Dimensionality Reduction**: Reducing the number of variables in a dataset.
- **Data Summarization**: Capturing key patterns while discarding noise.
- **Improving Interpretability**: Making data easier to visualize and analyze.

Why use PCA?

- PCA finds structure in large datasets by emphasizing **variation** and identifying **key patterns**.
- Widely used in **image processing**, **genomics**, and **natural language processing**.

PCA in Simple Terms

Goal

To efficiently represent data with fewer variables while **preserving the most important information**.

Example

Imagine we have three variables describing a person:

- x_1 = Height
- x_2 = Weight
- x_3 = Arm Length

PCA reduces these to 2 **principal components** that summarize these characteristics while preserving maximum variation.

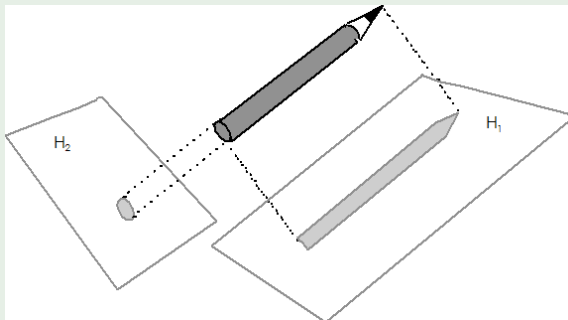
How Does PCA Help?

- **Data Visualization:** Reducing dimensions makes it easier to plot data and identify patterns.
- **Redundancy Elimination:** Removes redundant information when variables are strongly correlated (e.g., height and weight).

PCA graphically

When we want to represent a 3-D object in a paper, we do in that way that capturing the maximum amount of information.

Example



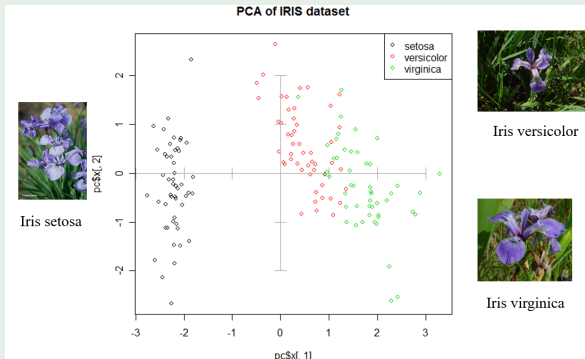
The image shows a the concept that we want to project on dimensions that **maximize the amount of variability**.

PCA Example: Iris Dataset

Example

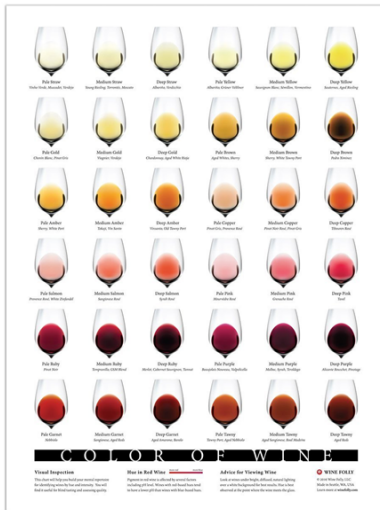
Consider a dataset with flowers of different species described by:

- Sepal length, Sepal width, Petal length, Petal width



This image shows a scatter plot of iris data, which PCA can help summarize by identifying the main axes of variation.

PCA Example: Wine Types



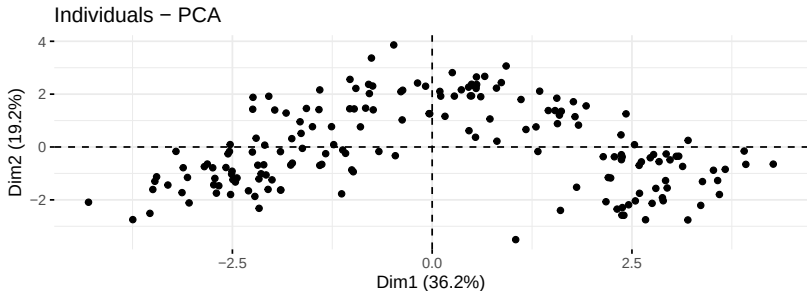
PCA Example: Wine Types

Data Example

Consider a dataset on **wines**, described by **Color Intensity**, **Alcohol Content**, **Age**, etc...

Example

PCA can summarize these correlated variables into a couple of dimensions that capture most of the variation, helping to classify different wine types.



Key Takeaways

Summary

- PCA transforms data to identify directions (or **principal components**) with maximum variance.
- Widely used in **AI and machine learning** to preprocess data, reduce noise, and make models faster and more interpretable.

Conceptual Foundation of PCA

Data Variability and Relationships

PCA explores data by finding the **directions of maximum variance**, revealing relationships between variables.

- PCA identifies patterns by examining **correlations** between variables.
- Variables that are strongly correlated indicate **redundant information**.
- The goal is to **reduce these correlations** to obtain a set of independent, informative components.

Example

Imagine a dataset containing measurements of height, weight, and arm length. These measurements are correlated and provide **redundant information** about **body size**. PCA identifies the main direction of variance, helping simplify the dataset.

Key Concepts in PCA

Principal Components

Principal components are **new variables** derived from the original data. They represent directions in which the data has the most variation, making them useful for summarizing complex datasets.

- **Principal components** are **linear combinations** of the original variables.
- Each component explains a **part of the total variance** in the data.
- The **first principal component** captures the **maximum possible variance**.
- These components are **independent**

What are linear combinations?

In PCA, a principal component is a combination of the original variables (e.g., height, weight) formed by adding them together with **specific weights**

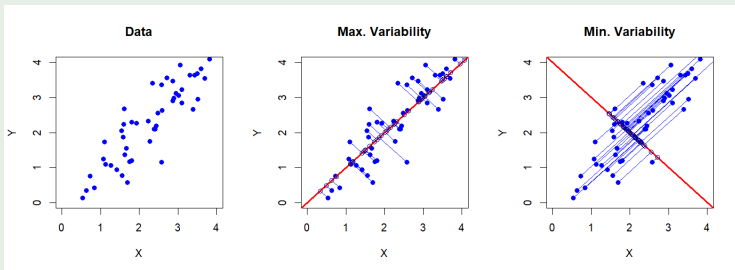
Visualizing Data with PCA

Example: Scatter Plot of Correlated Data

PCA finds a line that captures the direction with the highest variance.

- Data points aligned along a line indicate strong correlation.
- PCA rotates the data to align with this direction, allowing us to simplify and reduce dimensions.

Example



This scatter plot shows data that can be reduced to a single dimension, capturing most of the information in one principal component.

Main Steps: centering and scaling

Step 1: Centering the Data

The first step is to **center** the data by subtracting the mean of each variable from its values. This centers the data around the origin (zero mean) without altering the relationships between variables.

- Centering ensures that the first PC contains the **maximum variance**.
- By centering, we can focus on the **relationships between variables** rather than their absolute values.

Step 2: Scaling the Data (if necessary)

Scaling standardizes the range of each variable, especially important when the variables **have different units or scales**.

- Crucial step when some variables have much **larger ranges** than others.
- By scaling, **equal importance** to all variables is given.

When to Scale?

Scaling is typically applied when variables have **different units** (e.g., kilograms vs. meters) or vastly different ranges. In cases where all variables are measured on similar scales, scaling may not be necessary.

Main steps: Finding the Principal Components

Step 3: Calculating the Covariance Matrix

The **covariance matrix** shows the relationships between variables. Each element represents the covariance between a pair of variables, helping to understand how they vary together.

- Variables with **high covariance** are **strongly correlated**.
- This matrix is used to identify directions with **maximum variability**.

Step 4: Eigenvalues and Eigenvectors

Eigenvalues and **eigenvectors** help us find the **principal components** of the data.

- Each **eigenvalue** corresponds to a principal component's variance, indicating how much of the data's total variability it explains.
- **Eigenvectors** define the direction of each principal component.

Interpretation of Eigenvalues

The size of an **eigenvalue** tells us **how much of the total variance** is explained by its corresponding principal component. The largest eigenvalue indicates the component with the most variance.

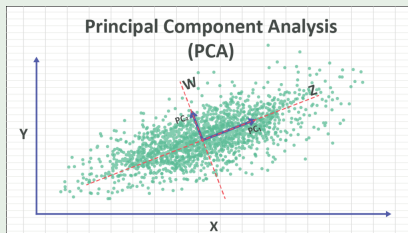
Main steps: Projecting the Data onto Principal Components

Step 5: Projecting Data onto Principal Components

The final step is to **project the original data** onto the new principal components. This creates a new dataset with reduced dimensions.

- Projecting onto the first few components captures most of the variability in the data.
- This reduces complexity while preserving important patterns.

Example



This image shows data projected onto the first two principal components, capturing the majority of its structure in two dimensions.

Summary of PCA Process

- 1 Center the data
- 2 Scale the data (if needed)
- 3 Calculate the covariance matrix.
- 4 Find eigenvalues and eigenvectors to determine principal components.
- 5 Project data onto the principal components to achieve dimensionality reduction.

Mathematical Background of PCA

The Covariance Matrix

The **covariance matrix** captures relationships between all pairs of variables. It's calculated as:

$$\text{Cov}(X) = \frac{1}{n-1} X^T X$$

where X is the centered data matrix. Each entry $\text{Cov}(X_{ij})$ represents how two variables vary together:

- A positive covariance indicates that two variables increase together.
- A negative covariance shows one variable increases as the other decreases.

Example

For example, if we calculate the covariance of height and weight, a high positive covariance would mean that as height increases, weight also tends to increase.

Eigenvalues and Eigenvectors

What are Eigenvalues and Eigenvectors?

Eigenvalues and eigenvectors are essential to finding the principal components of the data. For the covariance matrix $\text{Cov}(X)$:

- An **eigenvalue** represents the **amount of variance** captured by a principal component.
- An **eigenvector** defines the **direction** of each principal component in the data space.

In PCA, eigenvalues and eigenvectors satisfy the equation:

$$\text{Cov}(X) \cdot v = \lambda \cdot v$$

where:

- v is an eigenvector.
- λ is its corresponding eigenvalue.

Interpreting Eigenvalues and Eigenvectors

The largest eigenvalue corresponds to the direction (eigenvector) with the highest variance in the data. This is the first principal component. Each successive component captures less variance.

Understanding Variance Explained

Variance Explained by Principal Components

Each eigenvalue λ tells us how much of the total variance is captured by its corresponding principal component.

Proportion of variance explained by component i :

$$\frac{\lambda_i}{\sum \lambda}$$

This helps us determine how many principal components to keep.

Criteria to choose the number of components

Choosing the Number of Components

Several criteria:

- Typically, we select components that explain most of the total variance (e.g., **80-90%**) to simplify the dataset while preserving important patterns.
- **Kaiserrule**. Take all principal components with variance greater than the average variance.
- **Elbow rule**. The **screeplot** (we will see it later) helps to choose the number of components to take into account.
- **Statistical test** of significance of eigenvalues λ_α

Performing PCA in Practice: Steps

Step 1: Prepare the Data

To perform PCA, data should be **centered** (mean = 0) and often **scaled** to give each variable equal importance.

- Centering: Subtract the mean from each variable.
- Scaling: Divide by the standard deviation if variables have different units or ranges.

In R, this can be done using the `scale()` function:

```
data_scaled <- scale(data)
```

Example

As height (in cm) has higher values than weight, 'scale()' ensures both variables contribute equally by removing units and centering them around zero.

Calculating Principal Components in R

Step 2: Perform PCA in R

Use the 'prcomp()' function in R to calculate the principal components:
If you do not have scaled data, consider the next arguments:

- 'center = TRUE': Centers the data (mean = 0).
- 'scale = TRUE': Scales the data (standard deviation = 1).

```
pca_result <- prcomp(data, center = TRUE, scale = TRUE)
```

Output of 'prcomp()'

The 'prcomp()' function returns several key elements:

- 'sdev': Standard deviations of each principal component.
- 'rotation': The eigenvectors (loadings) that define each component.
- 'x': The data projected onto the principal components.

Interpreting PCA Results

Step 3: Analyzing the Variance Explained

Each principal component explains a portion of the data's total variance. This is indicated by the **eigenvalues**.

$$\text{Variance explained by component } i = \frac{\lambda_i}{\sum \lambda}$$

In R, the `summary()` function provides the proportion of variance explained:

```
summary(pca_result)
```

```
## Importance of components:
```

```
##
##          PC1      PC2      PC3      PC4      PC5
## Standard deviation  1.7361 0.8443 0.7529 0.64085 0.54373
## Proportion of Variance 0.6028 0.1426 0.1134 0.08214 0.05913
## Cumulative Proportion 0.6028 0.7454 0.8587 0.94087 1.00000
```

Example

In the example, we need 3 PC to retain at least the 80% of the variability.

Visualizing PCA Results

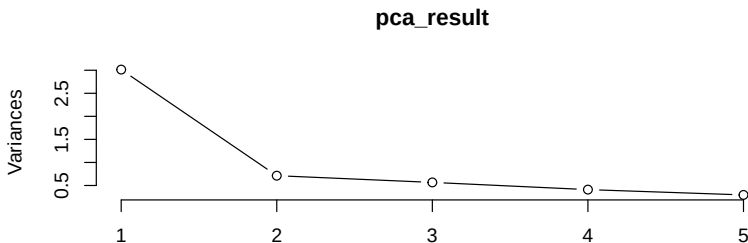
Step 4: Visualize the PCA Results

Visualization helps interpret PCA results. Common plots include:

- **Scree Plot:** Shows variance explained by each component.
- **Biplot:** Displays both variables and samples in the reduced space.

For a scree plot in R:

```
screeplot(pca_result, type = "lines")
```



Visualizing PCA Results with a Biplot

What is a Biplot?

A biplot is a visualization that represents both the observations (data points) and the variables (features) in the space of the first two principal components. It helps in understanding:

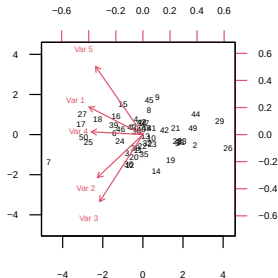
- **Observations (data points):** Shown as points, where proximity indicates similarity.
- **Variables (features):** Shown as arrows or vectors that indicate their contributions to each principal component.

Interpreting a Biplot

- **Data Points:** Points close together are similar in the reduced PCA space.
- **Variable Arrows:** The direction and length of each arrow show the influence of that variable on the components.
- **Angles Between Arrows:** Small angles between arrows indicate that the variables are positively correlated.

Creating a Biplot in R

Here is how to create a biplot in R using the `biplot()` function



Example

- All variables have same influence on PC1, but 3 and 5 have more influence on PC2.
- Variables 1, 2, 4 are correlated. Variables 3 and 5 are independent (90°).
- Individuals 26 and 29 are similar.

Calculating Variable Loadings in PCA

What are Variable Loadings?

Variable loadings represent the **contributions of each original variable** to the principal components. They show how much each variable influences the direction of each component.

- Loadings are the **coefficients** of the original variables in each principal component (i.e., they are the values in the eigenvectors).
- A **high loading** (positive or negative) indicates that the variable has a strong influence on the component.
- Loadings help in **interpreting** the principal components and understanding what each component represents in terms of the original variables.

Interpreting Variable Loadings

- Variables with **high loadings** on a component are strongly associated with that component.
- The **sign** (positive or negative) indicates the direction of the association.
- Loadings close to **zero** suggest that the variable has little impact.

Calculating and Extracting Loadings in R

R Code to Calculate Loadings

The loadings can be extracted from the 'rotation' attribute of a PCA object created with 'prcomp()':

```
(loadings <- pca_result$rotation)
```

##		PC1	PC2	PC3	PC4	PC5
##	[1,]	-0.4972219	0.2540520	-0.08857251	0.1681559	0.80753105
##	[2,]	-0.4200990	-0.3971136	-0.75719099	0.1699591	-0.25217712
##	[3,]	-0.3988612	-0.6179898	0.61952623	0.2713086	-0.03971384
##	[4,]	-0.4757316	0.0259715	0.09238657	-0.8673482	-0.11034828
##	[5,]	-0.4368940	0.6286274	0.16269198	0.3419594	-0.52014072

Example

In the output, observe each variable's loading on each principal component. For example, variables 3 and 5 have haigher impact on **PC2** (0.618 and 0.629, respectively).

Practical Example: Analyzing PCA Output

Using PCA Results in Practice

Let's apply these concepts to interpret PCA output in practice. We will:

- Review the explained variance to **decide the number of components**.
- Analyze the loadings to **interpret each component**.
- Use **biplots** to visualize relationships between variables and observations.

Guiding Questions for Interpretation

- Which components explain most of the variance?
- What do the loadings suggest about each component's meaning?
- How do observations cluster or separate based on the principal components?

Advices

- Analyze the variance explained to determine important components.
- Interpret loadings to assign meaning to each component.
- Use visualizations like scree plots and biplots to understand data patterns.

Practical Example of PCA: Iris Dataset

About the Iris Dataset

The 'iris' dataset in R contains measurements of 150 iris flowers from three species: setosa, versicolor, and virginica. The variables include:

- Sepal.Length
- Sepal.Width
- Petal.Length
- Petal.Width

Our goal is to use PCA to reduce the dimensionality of this dataset and observe the relationships between flower species.

Step 1: Preparing the Data

Data Preparation

First, we remove the species column and scale the data to give each variable equal weight in the PCA.

```
data <- iris[, 1:4]
data_scaled <- scale(data)
```

Example

Scaling centers each variable to have mean 0 and standard deviation 1, allowing PCA to capture the most variation across variables without bias.

Step 2: Performing PCA

Run PCA with 'prcomp'

We apply the PCA function 'prcomp()' on the scaled data:

```
pca_result <- prcomp(data,          center = TRUE, scale = TRUE)
pca_result <- prcomp(data_scaled)
```

Inspecting PCA Output

The 'summary(pca_result)' function provides the proportion of variance explained by each principal component.

```
summary(pca_result)
```

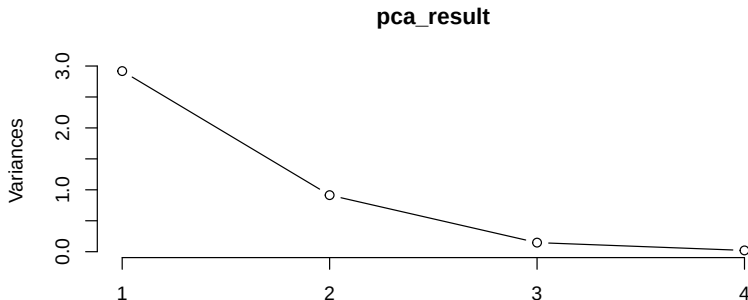
```
## Importance of components:
##
## Standard deviation      1.7084 0.9560 0.38309 0.14393
## Proportion of Variance 0.7296 0.2285 0.03669 0.00518
## Cumulative Proportion  0.7296 0.9581 0.99482 1.00000
```


Step 3: Analyzing the Variance Explained

Variance Explained

We look for the components that explain the majority of the total variance.

```
screeplot(pca_result, type = "lines")
```

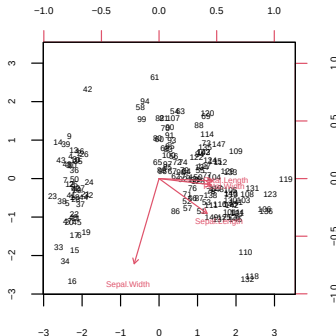


Example

The **first two components** explain most of the variance.

Step 4: Visualizing the Principal Components

We use a biplot to visualize both samples and variables in the first two PC.



Interpreting the Biplot

- Data points (flowers) close to each other have **similar measurements**.
- Vectors show the **influence** of each variable on the principal components.
- Different species may form **clusters**, indicating distinct characteristics.

Step 5: Practical Interpretation of Results

Interpreting PCA Components

Using the loadings and scores, we interpret the meaning of each component in relation to the original variables.

```
pca_result$rotation
```

##		PC1	PC2	PC3	PC4
##	Sepal.Length	0.5210659	-0.37741762	0.7195664	0.2612863
##	Sepal.Width	-0.2693474	-0.92329566	-0.2443818	-0.1235096
##	Petal.Length	0.5804131	-0.02449161	-0.1421264	-0.8014492
##	Petal.Width	0.5648565	-0.06694199	-0.6342727	0.5235971

Example

If the first component has high loadings for petal length and petal width, it may represent overall petal size. This helps us understand the main source of variation in the dataset.

Summary of PCA on Iris Dataset

What We Learned

By applying PCA to the 'iris' dataset:

- We have reduced the dimensionality from four to **two principal components**.
- We can see that **there are two main dimensions of characteristics** that retain more than 90% of the variability.
- Biplots show that one dimension is the **petal size** and the other dimension is the **sepal width**
- Visually, we identify **two/three main clusters** of data.

Including Supplementary Variables

What are Supplementary Variables?

Supplementary variables, also known as **passive variables**, are variables that are not used to create the principal components in PCA but are projected onto the components afterward.

- Supplementary variables allow us to **explore additional variables** without affecting the main PCA structure.
- These variables are often added to **analyze relationships with the components** or other active variables.
- They can be **quantitative or categorical** and offer insights into trends or groups within the data.

When to Use Supplementary Variables?

Supplementary variables are useful when:

- You want to add variables that **didn't contribute to the creation of the components**.
- You want to **explore relationships between additional variables and the principal components**.

Projecting Supplementary Variables onto PCA Components

How to Project Supplementary Variables

To project a supplementary variable onto existing principal components:

- For **quantitative supplementary variables**, calculate their coordinates by projecting them using the eigenvectors (loadings) of the components.
- For **categorical supplementary variables**, calculate the average position of each category within the component space.

In R, you can use packages like FactoMineR or factoextra for this purpose. In FactoMineR, use the `PCA()` function with the `sup.var` argument to add supplementary variables:

```
library(FactoMineR)
result <- PCA(data, quanti.sup = c(5,6), quali.sup = 7)
```

Summary of Principal Component Analysis (PCA)

Core Concepts of PCA

PCA is a technique for:

- **Dimensionality Reduction:** Reduces the number of variables, simplifying data while retaining key information.
- **Maximizing Variance:** Identifies directions in the data with the most variability.
- **Creating Uncorrelated Components:** Converts correlated variables into a set of independent components.

When to Use PCA

PCA is most useful when:

- You have **large datasets** with many variables.
- There is a need for **data simplification** and visualization.
- You want to **remove redundancy** by eliminating correlated variables.

Key Steps in Performing and Interpreting PCA

Performing PCA

- **Data Preparation:** Center and scale the data.
- **Calculate Components:** Use the covariance matrix to find principal components.
- **Interpret Components:** Examine variance explained to determine the importance of each component.

Interpretation and Visualization

- **Scree Plot:** Helps determine the number of components to retain.
- **Biplot:** Visualizes relationships between variables and observations in reduced dimensions.
- **Loadings and Scores:** Loadings indicate variable contributions, scores show data projected onto components.

References

- I.T. Jolliffe. *Principal Component Analysis*. Springer, 2nd Edition, 2002.
- Richard A. Johnson and Dean W. Wichern. *Applied Multivariate Statistical Analysis*. Pearson, 6th Edition, 2007.